	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Version 2E** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.	**Assessment criteria:** **C8**: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. **C9**: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative. **C10**: Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

1. Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.15	0.6	0.25
Essential (A_1)	120	105	78
Shared (A_2)	165	175	48
Extensive (A_3)	280	58	-42
Modular (A_4)	200	155	-22

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	120	78	160	100.5
A_2	175	48	115	141.75
A_3	280	-42	120	66.3
A_4	200	-22	100	117.5

(a) Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \le s \le 110$) increases the low-demand payoffs of A_3 and A_4 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of s. **[0.5]**

(c) Among eligible camera rental plans, find the smallest allowed subsidy s at which the Maximin recommendation changes from A_1 to A_4 alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \le s \le 110$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

2. Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.25	0.75
Portable (C_1)	112	76
Full-service (C_2)	180	26
Deluxe (C_3)	212	-6
Fixed (C_4)	164	56

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	112	76	100	85
C_2	180	26	50	64.5
C_3	212	-6	82	48.5
C_4	164	56	48	83

(a) Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A sponsor replaces C_1's large-audience payoff of 112 with g (integer, $112 \leq g \leq 186$). An alternative is eligible only if all its payoffs are at least 48. Determine each refreshment kiosk's eligibility based on the value of g. **[0.5]**

(c) Among eligible refreshment kiosks, find the smallest allowed sponsored payoff g at which the Minimax Regret recommendation changes from C_4 to C_1 alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $112 \leq g \leq 186$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

3. Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	184	80	78	155.4
D_2	210	54	52	173.6
D_3	158	106	104	139.8
D_4	236	28	78	173.6
D_5	132	80	130	113.8
D_6	262	2	104	173.6

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

4. Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	163	51	40	120.6
F_2	131	83	72	113.4
F_3	147	67	56	127.8
F_4	203	11	72	108.6
F_5	115	75	88	99
F_6	179	35	48	114.2

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

5. Art Materials Orders [C9-2]

An art materials distributor selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	196	118	117	167.4
H_2	222	92	91	188.2
H_3	274	40	78	168.7
H_4	170	105	143	144
H_5	313	1	117	159.6
H_6	248	66	65	179.1

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.